

Knowledge Journal

SVR

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1 Isolating CPU Cores

1. isolating CPU-cores

- "isolcpus" is the traditional, built-in method to isolate the cpu from the scheduler.
- essentially allows us to prevent the OS-scheduler from assigning any process or thread to our core.
- It remains idle until we assign a program/process to it explicitly using taskset or numactl
- To use this, open /etc/default/grub, and add isolcpus=2,3 (to isolate cpu 2 and 3) to GRUB_CMDLINE_LINUX_DEFAULT string
- GRUB_CMDLINE_LINUX_DEFAULT="quiet splash isolcpus=2". Save the file, exit.
- update the GRUB bootloader using "sudo update-grub"
- reboot the system for the changes to take place. sudo reboot.
- also isolcpus only prevents user-tasks from being assigned there. It might still use it for hardware interrupts.
- to prevent this, move the hardware interrupts to somewhere else, like core 0(Make sure to do this after rebooting).
- To do this, "echo 1 | sudo tee /proc/irq/default_smp_affinity"
- now that the core is just chilling, assign your binary executable to it in the following manner
- taskset -c 2 ./your-binary (Naturally, the 2 indicates the core to which you wanna write to this)

2. undoing the cpu-core isolation

- sudo nano /etc/default/grub
- remove the isolcpus=2 fom the GRUB_CMDLINE_LINUX_DEFAULT string
- sudo update-grub
- sudo reboot